

National University of Computer and Emerging Sciences, Lahore Campus			
	Course:	Embedded Systems	Course Code: EE419
	Program:	BS (EE)	Semester: Fall 2018
	Duration:	1 Hour	Total Marks: 30
	Paper Date:	2-Oct-18	Weight: 15 %
	Section:	ALL	Page(s): 1
	Exam:	Midterm - I	
	Name:		Roll Number: L15-4522
Instruction/Notes:		➤ The exam is closed book and notes. ➤ Use only black or blue pen to solve the paper. ➤ Use answer sheets to solve this exam. Please write in neat handwriting.	

Question # 1 Computer Architecture

CLO – 1 [10 marks]

- What are three main characteristics of an embedded system?
- List any two characteristics of RISC architecture
- Consider a 32 bit number 0x12131415. Store this number in byte addressable memory in big endian format. Memory addresses starts from address 0x2000.0850
- What are the two operating modes of ARM processor?

Question # 2 Programming ARM in C

CLO – 2 [20 marks]

- In an 8-bit variable **temp**, bit numbers 1 to 6 are flag bits. Write C code to change flag bits to "101010". All other bits should remain unchanged. [4 marks] —
- We want to access pin number 0 and 2 of PORTB. Use *bit specific addressing* method to — access these particular pins by defining a macro. Constant is defined as 4×2^b (b=bit number) and base address of PORTB is 0x4000.5000. [3 marks]
- ③ Name any two storage class specifiers. [3 marks] —
- ④ Write a code to implement a four bit counter using *read-modify-write* method. Assume four LEDs are connected with pins 2-5 of PORTB of TM4C controller. Counter should start from 0000 to 1111 and reset to 0000. Each value should be displayed for some time duration. [10 marks]

TM4C GPIO registers

Clock registers: SYSCTL_RCGCGPIO_R, SYSCTL_PRGPIO_R

Lock register: GPIO_PORTx_LOCK_R (unlock value = 0x4C4F434B) ➔

➔ Commit register: GPIO_PORTx_CR_R

Direction register: GPIO_PORTx_DIR_R ➔

Pull up register: GPIO_PORTx_PUR_R

Pull down register: GPIO_PORTx_PDR_R

➔ Digital enable register: GPIO_PORTx_DEN_R

Data register: GPIO_PORTx_DATA_R ➔

8 7 6 5 4 3 2 1 0
0 0 0 1 1 1 1 0